

error-correcting block code + message

rateless code (still error-correcting, but w/o rate)

4 indexing

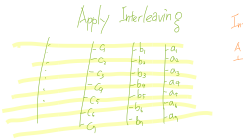
Traditional way

index/label/numbering

message/information

But Wait

DNA float



Interference

AAGGCCCTT

burst error/ CAAAAAT

SI

deletion

ACTGTA

ACGTA

substitution

ACTGTA

ACAGTA

insertion

ACTGTA

AACTGTA



The Ultimate Q

DNA coding

Can information last 1000 yr?

The Buffer Trick

a₁ a₂ a₃ a₄ a₅ a₆ a₇ a₈

↓ add buffer

a₁ a₂ a₃ 0000 a₄ a₅ 0000 a₆ a₇ a₈

↓ buffer

d₁ 0000 e₁ 0000 n

↓ buffer

d₁ 0000 e₁ 0000 n

Deletion channel

↓

Deletion channel

↓

Substitution channel

↓

Substitution channel

把 Burst 「平均」掉

让它看起来「独立同分布」!

I I D

Interleaving

-组

a₁ a₄ a₇

a₈ a₅ a₂

-组

a₉ a₆ a₃

2缺1

a₁ X a₁

a₈ X a₂

2缺1

d₉ X a₃

Burst error.

a representative model.

a₁ a₂ a₃ a₄ a₅ a₆ a₇ a₈

↓

a₁ a₂ a₃ X X X a₇ a₈

Burst error

Horizontal Thinking & Coding Theory

王新博

What is Coding?

Transmitting information from point A to point B

Transmitting information from time T to time U



Engineering

Mathematics